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Numerical Solutions of Spherical Blast Waves

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The strong-shock, point-source solution and spherical isothermal distributions were used as initial conditions for a numerical integration of the differential equations of gas motion in Lagrangean form. The von Neumann-Richtmyer artificial viscosity was employed to avoid shock discontinuities. The solutions were carried from two thousand atmospheres to less than one-tenth atmospheres peak overpressure. Results include overpressure, density, particle velocity, and position as functions of time and space. The dynamic pressure, the positive and negative impulses of both dynamic pressure and static overpressure, positive and negative durations of pressure and velocity, and shock values of all quantities are also described for various times and radial distances. Analytical approximations to the numerical results are provided.